Azure Hub-Spoke Enterprise Network Architecture

Professional Project Documentation

Project Overview

This project demonstrates an enterprise-grade Azure Hub-and-Spoke networking architecture. Shared networking and security services are centralized in a Hub Virtual Network while application workloads are isolated in a Spoke Virtual Network. The infrastructure is deployed using ARM Templates following Infrastructure as Code (IaC) practices.

Problem Statement

Flat cloud networks become difficult to secure and manage as environments grow. This solution provides centralized security, network segmentation, controlled routing, and repeatable deployments.

Solution Overview

The Hub VNet hosts Azure Firewall, Firewall Management, and Gateway subnets. The Spoke VNet hosts three isolated tiers: Web, Application, and Database. VNet Peering securely connects both virtual networks.

Key Features

- Hub-Spoke Architecture
- Azure Firewall
- VNet Peering
- Network Security Groups (NSGs)
- User Defined Routes (UDRs)
- ARM Templates (IaC)

Azure Services Used

Azure Virtual Network, Azure Firewall, Firewall Policy, VNet Peering, Network Security Groups, Route Tables (UDRs), Public IP Addresses, ARM Templates, Azure CLI.

Network Design

Hub VNet (10.1.0.0/16): AzureFirewallSubnet, AzureFirewallManagementSubnet, GatewaySubnet.
Spoke VNet (10.0.0.0/16): Web (10.0.1.0/24), Application (10.0.2.0/24), Database (10.0.3.0/24).

Security Design

Azure Firewall performs centralized outbound inspection while NSGs enforce communication only between required application tiers.

Traffic Flow

Internet → Azure Firewall → Hub VNet → VNet Peering → Web → Application → Database. Outbound traffic is redirected through UDRs to Azure Firewall before reaching the Internet.

Infrastructure as Code

ARM Templates automate infrastructure deployment, enabling consistent, repeatable, and version-controlled environments.

Deployment Process

1. Create Resource Group
2. Deploy ARM Template
3. Verify resources
4. Configure VNet Peering
5. Associate NSGs and Route Tables
6. Validate connectivity

Validation

Hub-Spoke connectivity, Firewall deployment, NSG rules, Route Tables, and ARM Template deployment were validated successfully.

Best Practices

Hub-Spoke topology, least privilege, layered security, centralized network management, and Infrastructure as Code.

Future Enhancements

Azure Bastion, Azure Monitor, Log Analytics, Azure VPN Gateway, Azure DDoS Protection, Application Gateway, Azure Front Door.

Conclusion

This project demonstrates a secure, scalable, and enterprise-ready Azure networking architecture aligned with Microsoft Cloud Adoption Framework networking recommendations.